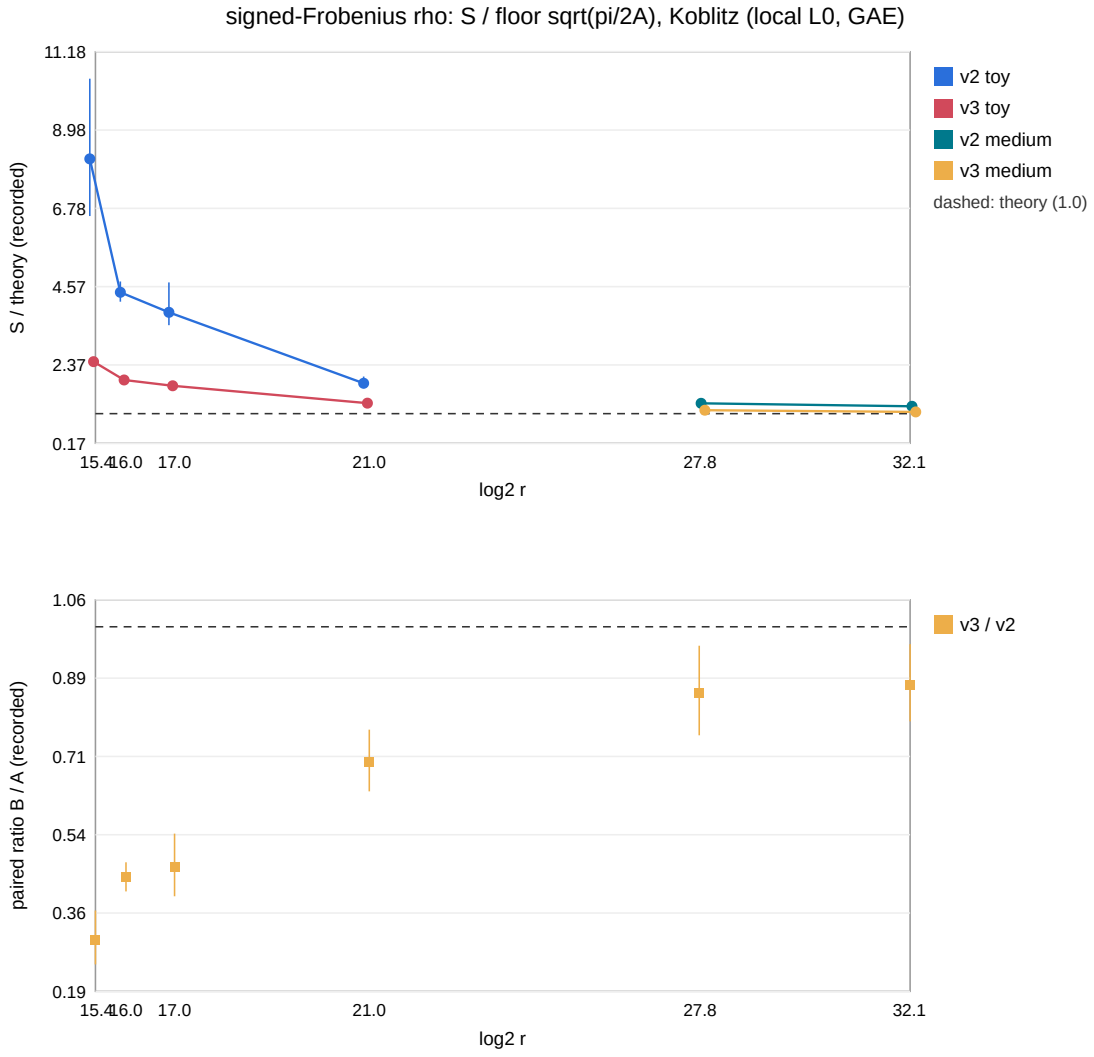
signed-Frobenius rho v3: the Koblitz walk at its floor

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Result (measured, counted GAE, local L0 host).

- `rho.signed_frobenius_v3` costs **0.459× `rho.signed_frobenius_v2` [0.418, 0.504]**, over 864 pairs on six Koblitz curves from 2^15.4 to 2^32.1. The order holds across the §5a band: 0.459 / 0.475 / 0.571 at w = 0 / 0.1 / 1.
- On the medium tier (2^27.8 and 2^32.1) v3 is **1.076× the floor √(π/2A) [0.998, 1.147]**; v2 is 1.249× [1.163, 1.333].
- Its search phase is 1.04×, 1.04× and 1.03× the floor (±0.045) at 2^21, 2^27.8 and 2^32.1.
- On the toy tier v3 is 1.876× the floor; v2 is 4.574×.



S / floor per tier (bound records) and the paired ratio per curve

What changed (commit `3592f4a172`)

The signed-Frobenius walk had the legacy shape: 8 jumps by joint multiplication and ⌊bits(r)/4⌋ distinguished-point bits. Those bits assume A = 2. With A = 2n the search is only √(πr/4n) steps. At 2^15.4 (n = 29, A = 58) that is about 30 steps, against a 16-step tail per walk.

`rho.signed_frobenius_v3` runs `rho_walk_v3` on the same classes with `RhoV3::frobenius()`:

- the ladder-built table, k ≈ √(F/2) jumps;
- distinguished-point bits round(½·$\log_2$F − 0.8), with F = √(πr/2A), fitted on the diagnostic: F = 30, 73, 268 gave best dp 2, 2, 3;
- the charged check [d]G reads its low bits from the set-up's G ladder.

The two options are off for `rho.negation_v3` and `rho.plain_v3`. A 600-run replay of that committed session reproduced identically after this change.

Measured values (S / floor; floor = √(π/2A))

curve	log2 r	A	v2 total	v3 total	v3 search ± SE	v3 set-up	v3 check	paired v3/v2 [95 %]
1:29	15.37	58	8.17	2.47	1.34 ± 0.05	0.74	0.39	0.302 [0.248, 0.368]
1:17	16.00	34	4.41	1.95	1.16 ± 0.05	0.57	0.22	0.442 [0.410, 0.475]
0:19	17.00	38	3.85	1.79	1.14 ± 0.05	0.45	0.19	0.464 [0.400, 0.539]
0:23	21.00	46	1.86	1.30	1.04 ± 0.04	0.19	0.07	0.699 [0.633, 0.771]
0:37	27.78	74	1.29	1.10	1.04 ± 0.04	0.05	0.01	0.852 [0.758, 0.958]
1:43	32.11	86	1.21	1.05	1.03 ± 0.05	0.02	0.003	0.871 [0.789, 0.962]

The totals come from `ecbench table`. The phase columns are derived as the mean recorded phase GAE / √r divided by the floor (`phase_constants.json`).

At 2^15.4 the whole floor is about 30 operations. The charged check [d]G alone (about 12 operations with the ladder) is 0.39 of it, and the set-up 0.74. These are O(log r) and O(k) and fall away with r, as the 2^21 to 2^32 rows show.

Run

item	value
spec	`research/ecbench_rho_v3_frobenius_20261107/spec.json`, `ECS1h1df58f076f8b`
session	`ECBS1h58ab29ffaaca`, complete; 3024 records, 3024 verified
commit / binary	`3592f4a172` / sha256 `5094bd45…f4a2`
host / isolation	Claude Code cloud container, **L0**; wall diagnostic only (ns/GAE v3/v2 1.12, within the A/A noise of 1.08)
audit	`audit.json`: `verify --replay-all`, 2592 replays reproduced, 0 problems
A/A	ops 1.0000 (deterministic)
bounds	toy `bound-candidate-toy.json` `ECBND1h83c2a9e181c5` (α 0.340, set-up dominated) and `bound-incumbent-toy.json` `ECBND1h0388eb67beda`; medium `bound-candidate-`

item	value
	`medium.json` ECBND1h536bc7c1f99e, `bound-incumbent-medium.json` ECBND1h5f63bae16e78 (two sizes: constant bounds)

Not established here

- No L2 isolation, no validator report: not a leaderboard or `benchmarks/README.md` row.
- The `_strong`, `_full` and `_wordram` Frobenius variants were not changed.